ToU ENGINE v0.3

Vector Representational Model

Initial Simulation Report

Extending ToU Engine from Scalar Dynamics to Structured Representational Regimes

Technical Supplement to the Akimov Theory of Unity Research Programme

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Methodological Note

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Abstract

This report introduces ToU Engine v0.3, a vector extension of the minimal (M_w, M_s) dynamical model in which the world-model and self-model are represented as n-dimensional real vectors rather than scalar values. The extension enables the system to stabilize structured configurations — patterns of coordinated values across dimensions — rather than single numerical equilibria. Initial simulation results at $n = 4$ demonstrate rapid perturbation recovery ($\tau_{relax} \approx 7$ steps), sustained inter-vector coordination (mean $\|M_w - M_s\| \approx 0.547$), and strong structural correlation ($r \approx 0.862$) between the world-model and self-model components. These results establish the existence of a structured joint regime: a stable configuration in which M_w and M_s maintain consistent relational organization without collapsing into identity. This regime is characterized as a pre-representational structural regime defined by stable relational organization across dimensions — a necessary precondition for representational capacity, not a demonstration of it. No claims are made regarding phenomenal consciousness, semantic content, or biological validity.

Keywords: vector model, structured regime, pre-representational structure, relational stabilization, perturbation recovery, joint coordination, dimensionality, ToU Engine.

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1. Motivation

ToU Engine versions v0.1 through v0.2c operated with scalar state variables: $M_w(t) \in R$ and $M_s(t) \in R$. In this setting, the system's dynamics converge to a single numerical equilibrium — a joint fixed point (M_w^*, M_s^*) in a two-dimensional state space. The scalar model suffices to demonstrate the existence of a perspective attractor and to characterize its stability and relaxation properties (v0.2b, v0.2c). It does not, however, permit the model to represent structured configurations: patterns, relations, or distributed encodings that vary across multiple dimensions simultaneously.

The ToU theoretical account of perspectival organization requires not merely that M_w and M_s converge to stable values, but that they maintain a coherent, coordinated representational relationship — a structured joint state in which the world-model and self-model mutually constrain one another across a range of dimensions. This requirement motivates the vector extension introduced in v0.3: by lifting M_w and M_s from scalars to vectors in R^n, the model gains the capacity to stabilize structured configurations rather than scalar equilibria.

ToU Engine v0.3 is not presented as a model of representation in the full philosophical sense. It is presented as a model of pre-representational structural stabilization: the emergence of stable, coordinated multi-dimensional joint states that constitute the minimal substrate from which representational capacity could arise, given appropriate functional organization.

> ***The system stabilizes structured configurations rather than scalar states. This represents a pre-representational structural regime: a necessary precondition for representational capacity, not a demonstration of it.***

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2. Model Extension: Scalar to Vector

The transition from the scalar to the vector model involves three structural changes:

2.1 State Space Dimensionality

In the scalar model, each state variable occupies a single dimension of the joint state space. The joint state $(M_w(t), M_s(t)) \in R^2$ evolves as a trajectory in a two-dimensional plane. In the vector model, each state variable occupies n dimensions:

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M_w(t) ∈ R^n
M_s(t) ∈ R^n
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The joint state (M_w(t), M_s(t)) now evolves as a trajectory in a 2n-dimensional state space. For n = 4, as used in the initial simulation, the joint state is a point in R^8. The attractor — if one exists — is a fixed point or bounded set in this higher-dimensional space.

2.2 Emergence of Dimensions

Each component M_w[i](t) and M_s[i](t) for i = 1,...,n represents a distinct dimension of the world-model and self-model respectively. These dimensions are not pre-assigned semantic content: in the minimal v0.3 implementation, they are undifferentiated real-valued channels. The significance of the vector extension lies not in the semantic interpretation of individual dimensions, but in the structural property that the system can now stabilize a pattern of values across dimensions — a configuration — rather than a single value.

2.3 From Value Stabilization to Configuration Stabilization

In the scalar model, attractor convergence means that M_w and M_s approach specific numerical values (M_w^*, M_s^*) and remain there under perturbation. In the vector model, attractor convergence means that M_w and M_s approach specific configurations ($M_w^* \in R^n$, $M_s^* \in R^n$) — structured patterns across n dimensions — and maintain their mutual coordination under perturbation. The system no longer stabilizes a state; it stabilizes a structure. This is the essential qualitative advance of v0.3 over its predecessors.

> *The vector extension is conservative in one important respect: it does not introduce new dynamical mechanisms. The same update rule — nonlinear element-wise update, memory accumulation, stochastic noise, bidirectional coupling — applies dimension-by-dimension, supplemented by cross-dimensional interaction through a weight matrix W. The qualitative novelty is structural, not dynamical.*

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3. Vector Dynamics

The update equations for ToU Engine v0.3 extend the scalar v0.2 dynamics to vector-valued state variables with cross-dimensional interaction:

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M_w(t+1) = M_w(t) + a·tanh(E(t)−M_w(t)) +
θ(M_s(t)−M_w(t)) + W·M_s(t) + α·M_w(t−1) + σε_w(t)
M_s(t+1) = M_s(t) + b·tanh(I(t)−M_s(t)) +
θ(M_w(t)−M_s(t)) + W·M_w(t) + α·M_s(t−1) + σε_s(t)
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where all variables are now vectors in R^n, all additions and subtractions are element-wise, and the following components are unchanged from v0.2:

- $a, b \in (0, 1)$: element-wise learning rates governing input tracking speed.
- $\theta \geq 0$: scalar coupling strength, applied element-wise to the difference ($M_s - M_w$) and ($M_w - M_s$).
- $\alpha \in [0, 1)$: memory coefficient, applied element-wise to the previous state vector.
- $\sigma > 0$: noise magnitude; $\varepsilon_w(t), \varepsilon_s(t) \in R^n$ are independent Gaussian noise vectors.

The new component introduced in v0.3 is the weight matrix W:

- $W \in R^{n\times n}$: cross-dimensional interaction matrix. The product $W \cdot M_s(t)$ adds to each dimension of M_w an affine combination of all dimensions of M_s, weighted by the corresponding row of W. This term introduces cross-dimensional coupling: the state of dimension i in M_w is influenced not only by the corresponding dimension i in M_s (via the element-wise coupling term θ), but also by all dimensions $j \neq i$ of M_s, according to the weights $W[i, j]$.

The weight matrix W governs the relational structure of the cross-dimensional interaction. In the initial v0.3 simulation, W is initialized as a random matrix with small entries and held fixed throughout the simulation. In subsequent extensions, W may be learned, structured, or sparse, permitting the study of different representational geometries.

> *The introduction of W marks the qualitative distinction between the scalar and vector models. Without W, the vector model reduces to n independent copies of the scalar model operating in parallel, each stabilizing its own dimension independently. With W, the dimensions interact: the system stabilizes a joint configuration across all n dimensions simultaneously. It is this joint stabilization that constitutes the pre-representational structural regime.*

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4. Simulation Setup

The initial simulation is conducted with the following configuration:

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Dimensionality            : n = 4 Coupling strength        : θ =
0.20  (within recommended range from v0.2b) Memory coefficient
: α = 0.050  (standard working value from v0.2b) Noise magnitude
: σ = 0.010 Learning rates           : a = b = 0.10 Weight matrix
W          : random, entries ~ Uniform(-0.05, 0.05), fixed
Initialization            : M_w(0), M_s(0) ~ Uniform(-1.0, 1.0)
per dimension External input E(t)       : constant vector, E =
[0.5, -0.3, 0.7, 0.1] Internal input I(t)       : constant vector,
I = [0.4, -0.2, 0.6, 0.3] Simulation length        : 500 time
steps Perturbation time        : t_p = 300 (after confirmed
convergence) Perturbation             : dimension 1 of M_w
displaced by +0.5 Runs per configuration   : 20
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Convergence is confirmed prior to perturbation by verifying that the element-wise variance of M_w(t) and M_s(t) across the preceding 50 time steps falls below threshold ε = 0.01 in all dimensions. The perturbation displaces a single dimension of M_w by a fixed magnitude, leaving the remaining dimensions and M_s undisturbed, and recovery is tracked across all dimensions simultaneously.

Three metrics are reported for each run:

- τ_relax: the time (in steps from t_p) for the joint state to return within recovery tolerance ε = 0.01 in all n dimensions and remain there for a sustaining window of W_s = 10 steps.
- ‖M_w − M_s‖: the Euclidean distance between M_w and M_s in R^n, averaged over the final 50 steps of the simulation (post-recovery). This measures the degree of maintained separation between the two state vectors — neither identity nor divergence, but stable relational offset.
- r(M_w, M_s): the Pearson correlation coefficient between the components of M_w and M_s, averaged over the final 50 steps. This measures the structural correspondence of the two vector configurations: a high correlation indicates that M_w and M_s maintain consistent relative structure across dimensions.

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5. Results

Results are reported as means ± standard deviations across 20 simulation runs with randomized initialization and noise seeds.

5.1 Perturbation Recovery (τ_relax)

The mean relaxation time following single-dimension perturbation is τ_relax ≈ 7 steps, with low variance across runs. This rapid recovery stands in marked contrast to the extended relaxation times observed in v0.2c at the stability boundary (δ_relax ≈ 85–167 steps for the scalar model at α = 0.10). The short τ_relax in v0.3 reflects the operating parameters (α = 0.05, θ = 0.20), which are well within the stable interior of the parameter space established in v0.2b — far from the stability boundary. The vector extension does not impair recovery speed; the multi-dimensional joint state recovers from single-dimension perturbation as efficiently as the scalar model recovers within the stable interior.

> *The rapid τ_relax of ≈ 7 steps is a property of the chosen parameter configuration, not a general property of the vector model. Configurations approaching the stability boundary (higher α, lower*

> *θ) would be expected to exhibit longer τ_relax in the vector model, consistent with the critical slowing down dynamics characterized in v0.2c.*

5.2 Inter-Vector Distance (‖M_w − M_s‖)

The mean Euclidean distance between M_w and M_s in the post-recovery equilibrium is ‖M_w − M_s‖ ≈ 0.547. This non-zero distance is a substantively significant result. It establishes that the joint attractor is not a state of identity (M_w = M_s, which would correspond to ‖M_w − M_s‖ = 0) but a state of structured offset: M_w and M_s occupy distinct but coordinated positions in R^n. The world-model and self-model are not merged into a single undifferentiated state; they maintain distinct configurations that stand in a stable relational relationship to one another.

Coordination is preserved without collapse into identity. This is the minimal relational property required for the ToU framework: M_w and M_s must be distinct (otherwise there is no world-model/self-model distinction) and coordinated (otherwise there is no joint attractor). The result ‖M_w − M_s‖ ≈ 0.547 confirms that both conditions are satisfied simultaneously in the vector regime.

5.3 Structural Correlation r(M_w, M_s)

The mean Pearson correlation between the components of M_w and M_s in the post-recovery equilibrium is r ≈ 0.862. This high positive correlation indicates that the two vector configurations are structurally similar: they maintain a consistent pattern of high and low values across dimensions, despite being separated in Euclidean space. M_w mirrors the structural organization of M_s without being identical to it.

This combination — non-zero distance with high structural correlation — is the computational signature of the pre-representational structural regime. The two vectors are distinct (separated), coordinated (correlated), and stable (low τ_relax). Together, these three properties characterize the joint attractor as a structured configuration in the ToU sense: not a scalar equilibrium but a relational pattern maintained across multiple dimensions.

5.4 Summary of Key Results

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τ_relax (perturbation recovery)    : ≈ 7 steps avg ||M_w - M_s||
(inter-vector)    : ≈ 0.547 correlation r(M_w, M_s)            :
≈ 0.862  Interpretation:   - Rapid recovery: system is well
within stable interior   - Non-zero distance: M_w and M_s remain
distinct   - High correlation: structural organization is shared
- Joint result: coordinated without identity
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6. Interpretation

6.1 From Dynamics to Structure

The scalar model (v0.1–v0.2c) established that the ToU mechanism can generate a stable joint dynamical regime: a state where M_w and M_s converge to coordinated equilibrium values and recover from perturbation. The vector model (v0.3) establishes that the same mechanism can generate a stable joint structural regime: a configuration where M_w and M_s maintain consistent relational organization across multiple dimensions. The transition from v0.2 to v0.3 is therefore not merely a quantitative extension (more dimensions) but a qualitative one: the system's attractor is no longer a point in R^2 but a structured configuration in R^{2n}.

The model operates at the level of relational stabilization. The joint attractor is not defined by specific values but by specific relations: the pattern of differences, correlations, and cross-dimensional dependencies that M_w and M_s maintain relative to one another. It is this relational structure that is stable and recoverable — not any particular scalar value.

The system operates at the level of relational stabilization, where the attractor is defined not by fixed values but by stable relations across dimensions.

6.2 The Pre-Representational Structural Regime

The results establish the existence of what this report terms a pre-representational structural regime: a stable, multi-dimensional joint state in which M_w and M_s maintain consistent relational organization without external enforcement. This regime is 'pre-representational' in a specific sense: it provides the structural substrate that would be required for representational capacity but does not, by itself, constitute representation.

> A representation may be understood as a stable structured configuration within M_w, maintained through coupling with M_s. In the v0.3 vector model, such configurations arise and are maintained under perturbation. Whether they constitute representations in a semantically loaded sense depends on functional organization that lies beyond the scope of this minimal implementation.

This definition is deliberately operational: it identifies what the model produces (stable structured configurations) without committing to strong claims about what those configurations mean or whether they carry semantic content. The value of the pre-representational structural regime is that it is empirically tractable: it can be measured (through τ_{relax}, $\|M_w - M_s\|$, and r), it is reproducible across runs, and it provides a clear target for future extensions.

6.3 Structural Correspondence and Relational Identity

The high structural correlation ($r \approx 0.862$) between M_w and M_s reflects a deeper property of the vector attractor: the world-model and self-model develop and maintain a shared relational structure. The world-model's internal organization mirrors the self-model's, not because they are identical (they are not: $\|M_w - M_s\| \approx 0.547$) but because the bidirectional coupling ensures that each model's dimensional structure is shaped by the other's. This mutual structural constraint is the vector-level analogue of the reflexive stabilization described in the ToU theoretical account.

6.4 Relation to the ToU Hard Core

The vector model preserves and extends the three-layer structure of the ToU hard core. Layer 1 (functional integration) is extended: information is now integrated across n dimensions rather than a single scalar channel. Layer 2 (global coordination) is extended: coordination now encompasses the relational organization of entire vector configurations, not merely the alignment of scalar values. Layer 3 (reflexive stabilization) is unchanged in principle: the bidirectional (M_w, M_s) coupling continues to operate as the constitutive stabilizing mechanism. The perspective attractor in the vector model is a structured joint configuration in R^{2n}, with τ_{relax}, inter-vector distance, and structural correlation as its measurable properties.

> ***ToU Engine v0.3 demonstrates that the ToU coupling mechanism — unchanged in principle from v0.1 — is sufficient to stabilize not merely scalar equilibria but structured multi-dimensional configurations. The system stabilizes structured configurations rather than scalar states. This establishes the mechanistic basis for a pre-representational structural regime within the ToU framework.***

7. What This Does Not Prove

> **The following limits apply to all results reported in this document. They are methodologically constitutive, not incidental.**

The vector model results and pre-representational structural regime described in this report do not demonstrate, prove, or suggest any of the following:

- Phenomenal consciousness or subjective experience. A stable, high-dimensional joint configuration is a dynamical property of a computational system. It is not evidence of inner experience, qualia, or any first-person phenomenal state. The hard problem of consciousness is not addressed.
- Semantic content or intentionality in any philosophically substantive sense. The dimensions of M_w and M_s carry no pre-assigned meanings. The correlation $r \approx 0.862$ measures structural similarity between numerical arrays, not semantic correspondence between meaningful representations. The description of the joint state as 'structured' refers to its multi-dimensional organization, not to any semantic structure it might encode.
- Representation in the full philosophical or cognitive-scientific sense. The definition proposed in Section 6.2 is deliberately minimal and operational: a stable structured configuration within M_w maintained through coupling with M_s. This is a necessary condition for representation, not a sufficient one. Full representation would require, in addition, functional role, causal covariance with external states, and inferential integration — none of which are claimed here.
- Biological validity or neural correlates. No claim is made that the vector dimensions correspond to neural populations, cortical columns, or any biologically identifiable processing unit. The mapping between the abstract vector model and specific neural architectures is a task reserved for empirical extensions.
- Progress beyond the scalar model with respect to consciousness or sentience. The transition from scalar to vector does not bring the model closer to consciousness in any sense that the ToU framework would endorse. It brings the model closer to the structural conditions that a representational system would need to satisfy, which is a different and more modest claim.

The pre-representational structural regime is a computational property of the v0.3 vector model. It is not evidence of consciousness, representation, or biological validity. It establishes a structural substrate that is compatible with representational capacity, not a demonstration of it.

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8. Implementation Note

The initial v0.3 results establish the existence and basic properties of the pre-representational structural regime. The following studies are identified as the immediate next-step priorities. These studies aim to evaluate whether the structured regime exhibits pattern retention, relational recovery, and input-dependent stability — properties that would indicate a transition toward a proto-semantic regime.

- Pattern retention test: Present a specific input pattern (a structured vector E(t) with a defined signature) to the system during the convergence phase, then remove the pattern and observe whether the joint state retains a measurable trace of the pattern in its post-pattern configuration. This tests whether the vector attractor encodes input-specific information.

- Cross-dimension propagation: Apply a perturbation to a single dimension of M_w and track how the disturbance propagates across the other dimensions before equilibrium is restored. The propagation pattern reveals the cross-dimensional coupling structure induced by W and characterizes the system's internal dependency structure.
- Similarity metrics: Introduce cosine similarity as a complement to Pearson correlation for characterizing the structural relationship between M_w and M_s. Cosine similarity is scale-invariant and may provide a cleaner measure of directional alignment in higher-dimensional state spaces.
- Higher-dimensional systems: Extend the simulation to $n = 8$, 16, and 32 dimensions and characterize how τ_{relax}, $\|M_w - M_s\|$, and r scale with dimensionality. Understanding the scaling behaviour is necessary to assess the model's applicability to more realistic representational architectures.
- Multi-agent extension: Introduce two coupled vector (M_w, M_s) systems and study the emergence of shared structural configurations between agents. This extension tests whether the ToU coupling mechanism can support the formation of inter-agent relational structures — a necessary step toward a theory of shared perspectival organization.

9. Conclusion

ToU Engine v0.3 extends the ToU simulation programme from scalar dynamics to structured representational regimes. The transition from $M_w, M_s \in R$ to $M_w, M_s \in R^n$ introduces the capacity for configuration stabilization: the system no longer converges to a single numerical equilibrium but to a structured multi-dimensional joint state in which the world-model and self-model maintain consistent relational organization across n dimensions.

Initial simulation results at $n = 4$ confirm the existence and stability of the pre-representational structural regime: rapid perturbation recovery ($\tau_{relax} \approx 7$ steps), stable inter-vector separation ($\|M_w - M_s\| \approx 0.547$), and strong structural correlation ($r \approx 0.862$). These results establish that coordination is preserved without collapse into identity: M_w and M_s remain distinct, structurally similar, and jointly stable.

The v0.3 extension is best understood as a bridge between two phases of the ToU Engine programme. Versions v0.1–v0.2c established the dynamical foundations: stability, relaxation, and phase-transition behaviour. Version v0.3 initiates the structural phase: the capacity to stabilize configurations that are candidates for representational content. The next steps — pattern retention, cross-dimensional propagation, and similarity metrics — will test whether the structural regime exhibits the input-sensitivity and informational specificity required for a model of representation in the cognitive-scientific sense.

> ***ToU Engine v0.3 demonstrates that the ToU mechanism — bidirectional (M_w, M_s) coupling under nonlinear, memory-augmented dynamics — is sufficient to stabilize structured multi-dimensional configurations. This is a pre-representational structural result, not a claim about consciousness. It represents the first step in extending the ToU framework from a theory of dynamical organization to a theory of structured representational capacity, and establishes a bridge toward the broader project of the Akimov Theory of Everything.***

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